described it in 1961. Since then, the theory has been applied to many sciences, including population dynamics.

Chaotic populations

In the 1970s, Australian scientist Robert May became interested in animal population dynamics, and worked on a model to forecast growth or decline over time. This led him to the logistic equation. Devised by Belgian mathematician Pierre-François Verhulst, this equation produces an S-shaped curve on a graph—showing population growing slowly at first, then rapidly, before tapering off into a state of equilibrium.

May experimented with Verhulst's formula to create the "logistic map," which showed the population trends on a graph. Although it created predictable patterns at the lowest rates of growth, May found that the logistic equation produced erratic results when the growth rate was equal to or above 3.9. Instead of producing repeating patterns, the map plotted trajectories that appeared completely random. May's work showed how a simple, constant equation could produce chaotic behavior. His logistic map is now used by demographers to track and predict population growth.

"Chaos: when the present determines the future, but the approximate present does not approximately determine the future."

Edward Lorenz

See also: Predator–prey equations • Non-consumptive effects of predators on their prey • The Verhulst equation • Metapopulations